Mid Term Examination (March-2025)
B.Tech.(CSE/IT/CST/ITE)

Subject: Programming in Python | **Subject Code: CIE-332T**
Time: 1.30 Hrs | **Max. Marks = 30**

Note: Attempt three questions including Ques 1 which is compulsory

Q1. What would be output of the following code, Explain your answer: **2x5=10 Marks**

Choice 1

a)
```
import sys
def check(x):
    if x % 2 == 0:
        sys.exit()
    return x * 2
print(check(5) + check(3))
```
(CO1)

b)
```
x = 10
y = 20
z = (x < y) and (y > 15) or (x == 10) and not (y == 20)
print(z)
```
(CO1)

c)
```
def func(a, b=5):
    return a * b
x = func(3)
y = func(2, 10)
print(x + y)
```
(CO1)

d)
```
def compute(x, y=5, z=10):
    return x * y + z
result1 = compute(2)
result2 = compute(3, 4)
result3 = compute(1, z=20)
print(result1, result2, result3)
```
(CO2)

e)
```
word_list = ["hello", "world", "python"]
def modify_word():
    global word_list
    word_list[1] = word_list[1].upper()
    local_list = word_list[:2]
    return "-".join(local_list)
print(modify_word())
print(word_list)
```
(CO2)

OR

Choice 2

a)
```
x = "Automate"
y = "Python"
z = (x[:4] + y[-3:]) * 2
print(z[::3] + z[-2] + z[5])
```
(CO1)

b)
```
def test(n):
    if n % 2 == 0:
        return "Even"
    elif n % 3 == 0:
        return "Divisible by 3"
    return "Odd"
print(test(9) + " " + test(12))
```
(CO1)

c)
```
import pprint
data = {'name': 'Alice', 'age': 25, 'city': 'New York'}
pprint.pprint(data)
```
(CO2)

d)
```
d = {1: 'one', 2: 'two', 3: 'three'}
keys_list = list(d.keys())[::-1]
values_list = list(d.values())[::-1]
print(keys_list[0], values_list[1])
```
(CO2)

e)
```
data = {'Alice': 25, 'Bob': 30, 'Charlie': 35}
try:
    age = data['David']
except KeyError:
    age = sum(data.values()) // len(data)
print(age)
```
(CO2)

Q2. a. Explain the difference between local and global scope in Python. How does the ***global*** statement affect variable scope? Provide an example where modifying a global variable inside a function leads to unexpected behavior. **3 Marks (CO2)**